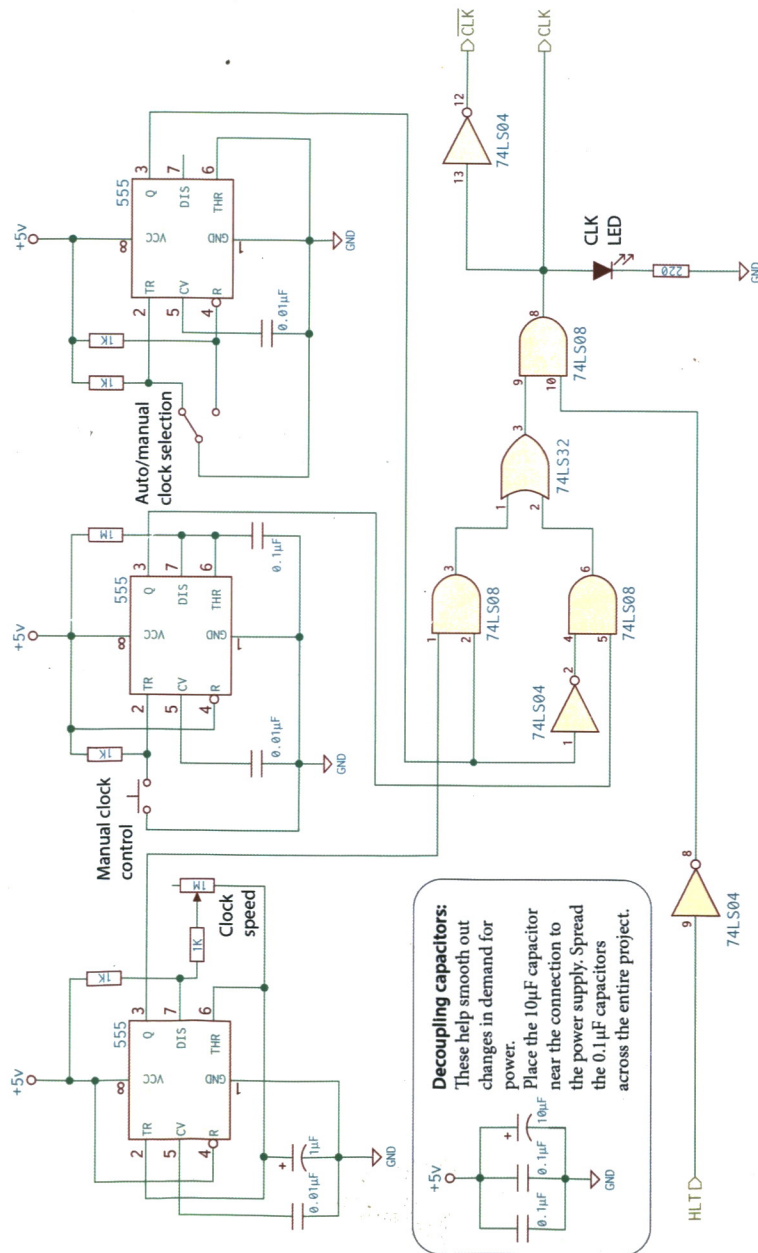


Clock module schematic



If you get stuck

There's no denying it: Building a computer from scratch on breadboards is a big project. The best part is that by the time you're done, you'll have an incredible understanding of how the computer works. But at least some of that understanding will come because you had to track down some (possibly frustrating) problems. The good news is you're now part of a community of people working on this project and helping each other out. There are several good places to look for help:

- Comment threads on the YouTube videos.
- Comment threads at eater.net/8bit.
- Questions and discussions at reddit.com/r/beneater.



Caution:

You should never connect LEDs directly to a 5 volt source without a current limiting resistor. Doing so will damage the LED.

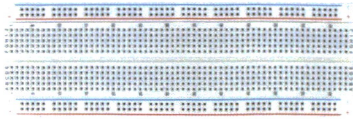
The outputs of most 74LSxx logic chips have internal resistors that let you to connect LEDs directly to the outputs without a resistor. (The 555 timer, 74189 RAM, 28C16 EEPROM, and 74HC595 *do not*.)

For the most reliable results, you may want to connect a 220Ω resistor in series with every LED anyway. I omitted many of these as a shortcut in the videos, but I've included enough 220Ω resistors in these kits if you choose (or need) to use them.

Components included in this kit

Note that some components might look slightly different than shown here or in the videos.

1 Breadboard



1 Jumper wire kit



The kit includes 140 pre-cut wires.

1 Power plug



5 volt, 2 amp

1 Power plug adapter



See below for how to use the adapter.

3 555 IC timers



1 74LS04 hex inverter



1 74LS08 quad AND gate



1 74LS32 quad OR gate



1 Momentary pushbutton



1 1MΩ potentiometer



1 Slide switch



This replaces the pushbutton toggle switch used in the videos.

5 220Ω resistors (red-red-brown)



Includes 4 extra

10 1kΩ resistors (brown-black-red)



Includes 5 extra

5 1MΩ resistors (brown-black-green)



Includes 4 extra

5 Yellow LEDs



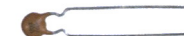
Includes 2 extra

5 Blue LEDs



Includes 3 extra

5 0.01μF capacitors (103)



Includes 2 extra

5 0.1μF capacitors (104)



Includes 2 extra

5 1μF capacitors



Includes 4 extra

5 10μF capacitors



Includes 4 extra

Powering your project

The power supply provides 5 volts and will power the entire computer. You can use the included adapter to easily connect it to your breadboards.

